

CO-1-AT1

OSI Layer Calculations and Network Communication Analysis

Assignment Submission-1

Course: Computer Networks

Topic: Application of OSI layers in enabling reliable data transmission across networks

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Question 1 – Transport and Data Link Layer Analysis

Introduction

The Transport Layer divides a large file into smaller segments for reliable transmission. The Data Link Layer encapsulates each segment into a frame by adding headers and trailers for node-to-node communication and error detection.

Given

- File Size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment Size = **1500 bytes**
- Data Link Layer Overhead = **40 bytes/frame**

Calculation

Step 1: Convert file size into bytes

$$150 \text{ MB} = 150 \times 1024 \times 1024$$

$$= \mathbf{157,286,400 \text{ bytes}}$$

Step 2: Number of segments

$$\text{Segments} = 157,286,400 \div 1500$$

$$= \mathbf{104,858 \text{ segments}}$$

Since one segment becomes one frame,

$$\text{Frames} = \mathbf{104,858}$$

Step 3: Data Link Layer Overhead

$$\text{Overhead} = 104,858 \times 40$$

$$= \mathbf{4,194,320 \text{ bytes}}$$

Step 4: Total transmitted data

$$157,286,400 + 4,194,320$$

$$= \mathbf{161,480,720 \text{ bytes}}$$

Convert to bits

$$161,480,720 \times 8$$

$$= \mathbf{1,291,845,760 \text{ bits}}$$

Step 5: Transmission Time

Transmission Time

$$= 1,291,845,760 \div 50,000,000$$

$$= \mathbf{25.84 \text{ seconds}}$$

Final Answer

- Number of Segments = **104,858**
- Number of Frames = **104,858**
- Data Link Layer Overhead = **4,194,320 bytes**

- Transmission Time = **25.84 seconds**

Explanation

The Transport Layer performs segmentation, sequencing, acknowledgements, flow control, and error recovery to ensure reliable communication. The Data Link Layer performs framing, adds headers and trailers, detects errors using CRC, and provides reliable node-to-node delivery.

Question 2 – Sliding Window Protocol

Introduction

The Sliding Window Protocol allows multiple frames to be transmitted before receiving acknowledgements, improving network efficiency.

Given

- Window Size = **6 frames**
- Total Frames = **48**
- User Data per Frame = **1024 bytes**

Calculation

Transmission Windows

$$48 \div 6$$

$$= \mathbf{8 \text{ windows}}$$

One frame is lost in every window.

Retransmissions

$$= \mathbf{8 \text{ frames}}$$

Final Answer

- Transmission Windows = **8**

- Retransmissions = **8**

Explanation

Flow control prevents the sender from transmitting data faster than the receiver can process it. It avoids congestion, minimizes packet loss, and improves communication efficiency.

Question 3 – IP Fragmentation

Introduction

When an IP packet exceeds the MTU, the Network Layer fragments it into smaller packets so that it can travel through the network.

Given

- IP Packet Size = **12,000 bytes**
- MTU = **1500 bytes**
- IP Header = **20 bytes**

Calculation

Payload per Fragment

$$1500 - 20$$

$$= \mathbf{1480 \text{ bytes}}$$

Fragments

$$12,000 \div 1480$$

$$= \mathbf{8.11}$$

Rounded up

$$= \mathbf{9 \text{ fragments}}$$

Header Overhead

$$9 \times 20$$

= 180 bytes

Final Answer

- Fragments Created = **9**
- Header Overhead = **180 bytes**

Explanation

The Network Layer performs logical addressing, routing, fragmentation, and reassembly. The destination device reassembles all fragments to reconstruct the original packet.

Question 4 – Sliding Window Protocol

Introduction

Sliding Window Flow Control improves network utilization by allowing multiple frames to be transmitted before acknowledgements are received.

Given

- Window Size = **6 frames**
- Total Frames = **48**

Calculation

Transmission Windows

$$48 \div 6$$

= 8 windows

Retransmissions

One frame lost per window

= 8 retransmissions

Final Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents receiver buffer overflow, improves throughput, reduces unnecessary waiting time, and ensures reliable communication.

Question 5 – Session Layer Synchronization

Introduction

The Session Layer establishes, manages, and synchronizes communication sessions. Synchronization checkpoints help resume communication after failures.

Given

- Video File = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Calculation

Checkpoints Created

$$390 \div 60$$

$$= \mathbf{6.5}$$

Completed checkpoints

$$= \mathbf{6}$$

Checkpoint Positions

- 60 MB
- 120 MB
- 180 MB

- 240 MB
- 300 MB
- 360 MB

Data to Retransmit

390 – 360

= **30 MB**

Final Answer

- Checkpoints Created = **6**
- Data to Retransmit = **30 MB**

Explanation

Synchronization checkpoints reduce retransmission time by allowing communication to resume from the last checkpoint instead of restarting the entire file transfer.

Conclusion

The OSI model provides a structured framework for reliable, secure, and efficient communication between networked devices. Each layer performs a specific function, from providing user services at the Application Layer to transmitting data through the Physical Layer. In this assignment, calculations related to segmentation, sliding window protocol, IP fragmentation, synchronization, compression, encryption, protocol overhead, transmission delay, and data efficiency demonstrate the practical application of OSI layer concepts. Features such as flow control, error detection, routing, fragmentation, synchronization, compression, and encryption work together to ensure accurate and reliable data transmission across modern computer networks. Understanding these concepts helps in designing, implementing, and troubleshooting efficient networking systems.

References

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